Exercise 8: Text Generation Using Hugging Face Inference API

Aim

To develop a Python program that uses a Hugging Face model through the Inference API to generate text from user-provided prompts and display the generated output.

Algorithm

1. Create an account on Hugging Face.
2. Generate an access token with permission to use Inference Providers.
3. Install the required Python library.
4. Import the InferenceClient class.
5. Connect to the Hugging Face Inference API using the token.
6. Accept a text prompt from the user.
7. Send the prompt to the model and display the generated response.

PROGRAM

```
from huggingface_hub import InferenceClient

token = input("Enter your Hugging Face token: ")

client = InferenceClient(api_key=token)



while True:

    prompt = input("\nEnter your prompt (or 'exit' to stop): ")

    if prompt.lower() == "exit":

        print("Program terminated.")
```

```
        break
    try:
        response = client.chat.completions.create(
            model="openai/gpt-oss-120b",
            messages=[
                {"role": "user", "content": prompt}
            ],
            max_tokens=200
        )
        print("\nGenerated Output:")
        print(response.choices[0].message.content)
    except Exception as e:
        print("Error:", e)
```

SAMPLE OUTPUT



RESULT

Thus, the Python program was developed successfully to accept user prompts, send them to a Hugging Face model through the Inference API, and display the generated text responses.